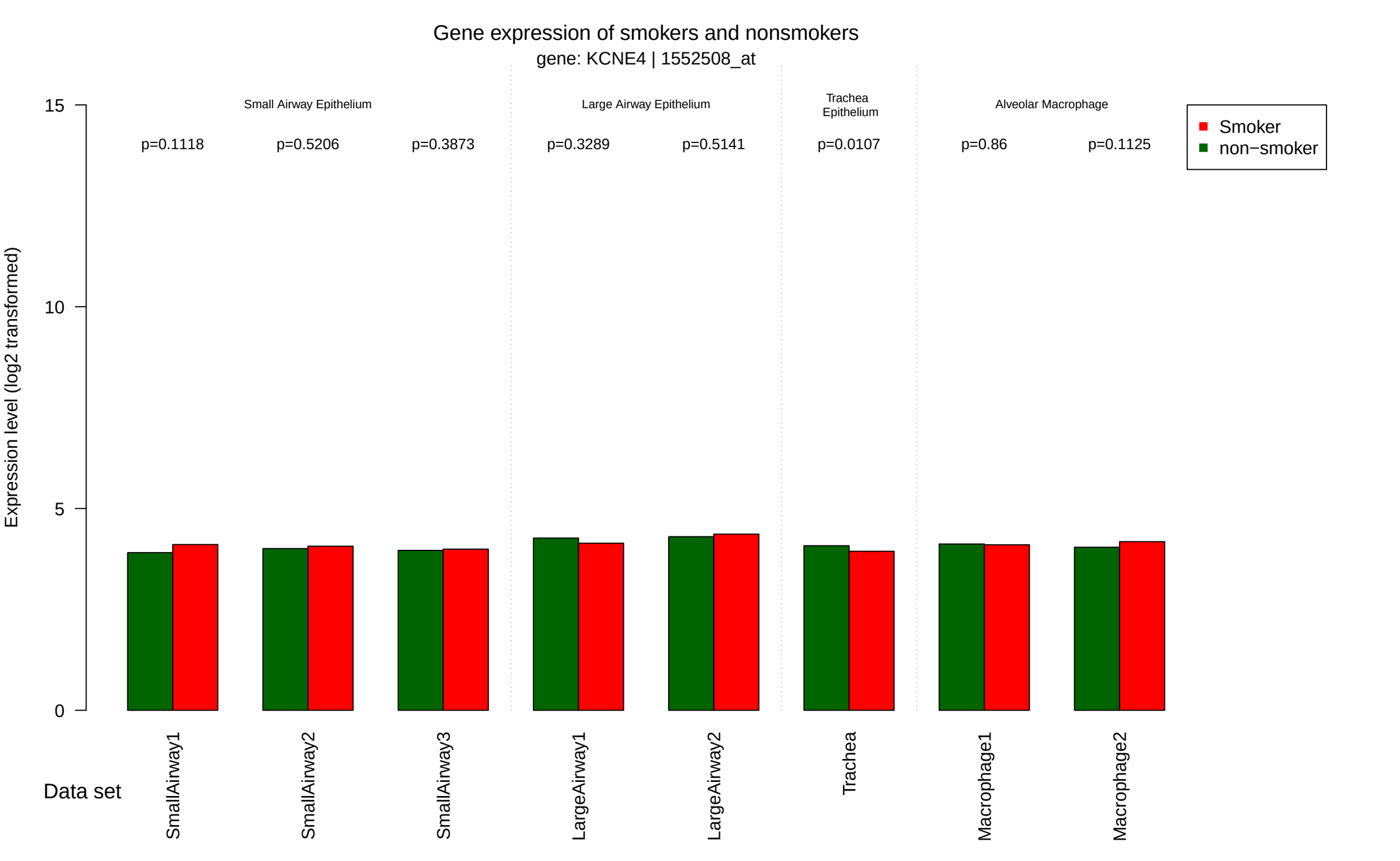
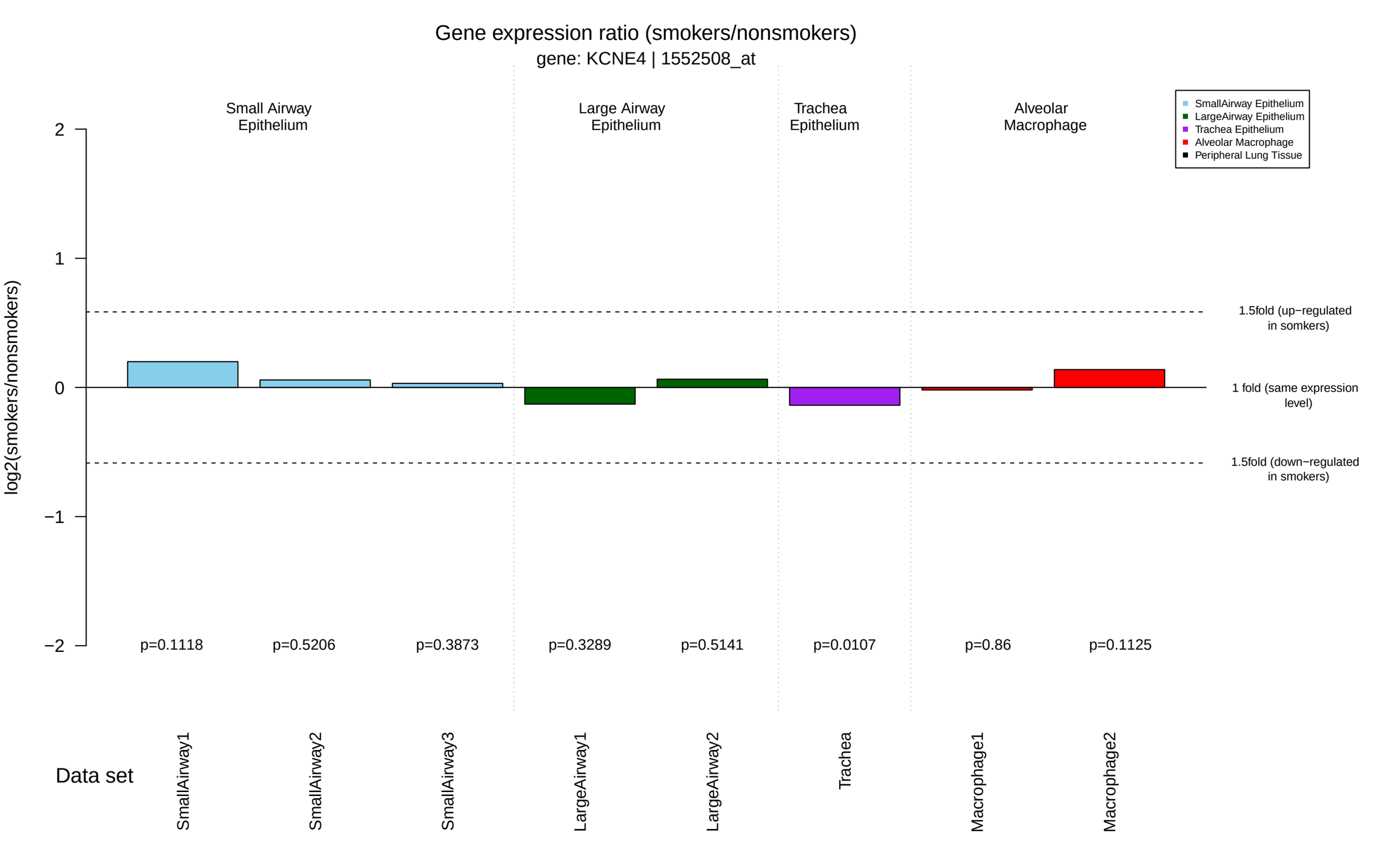
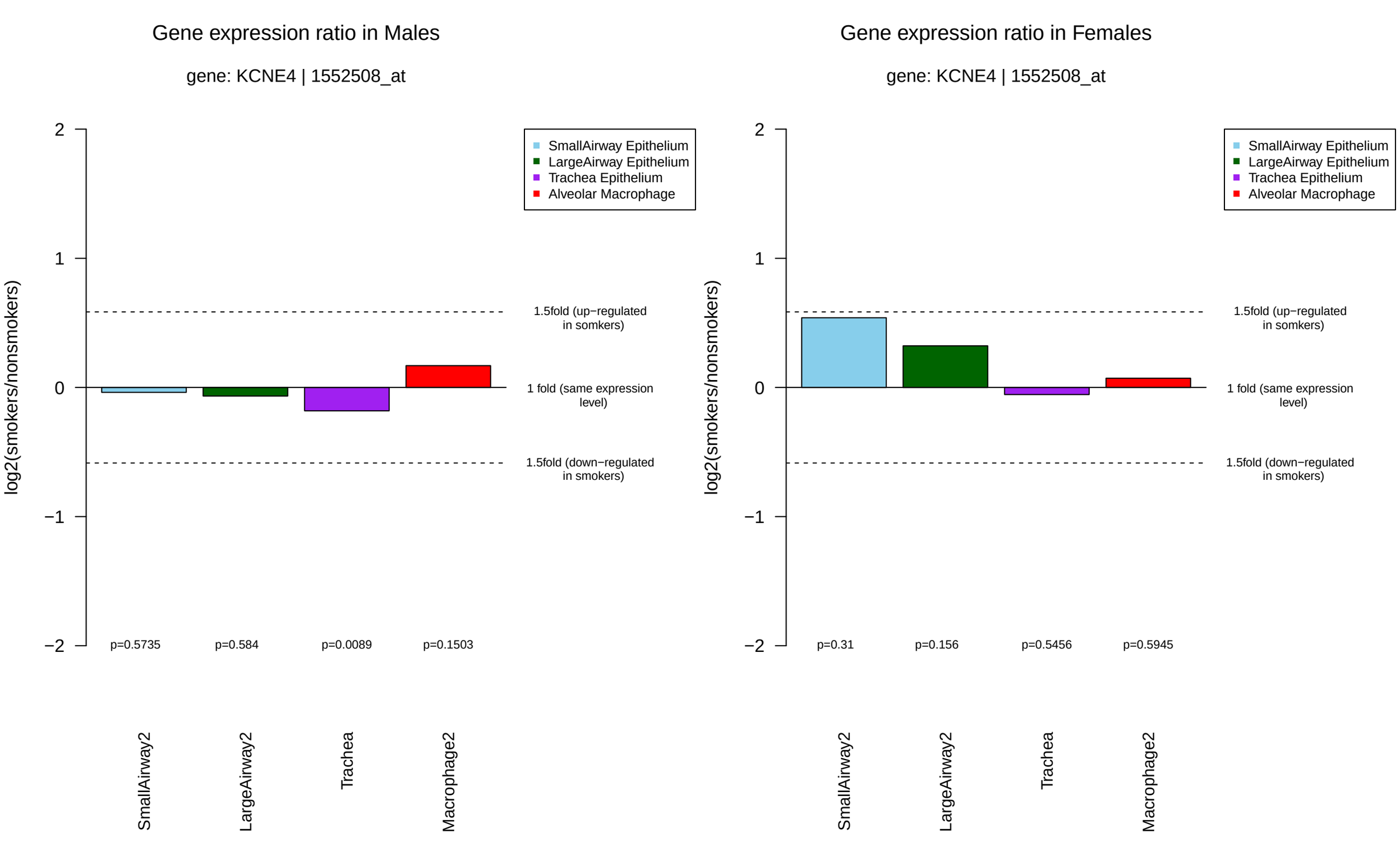
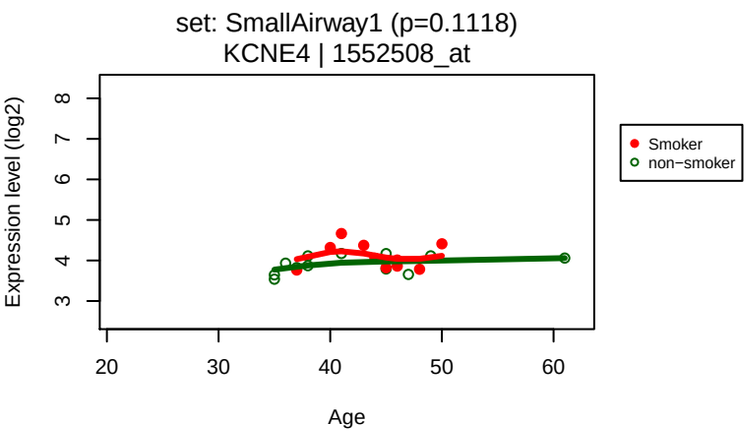
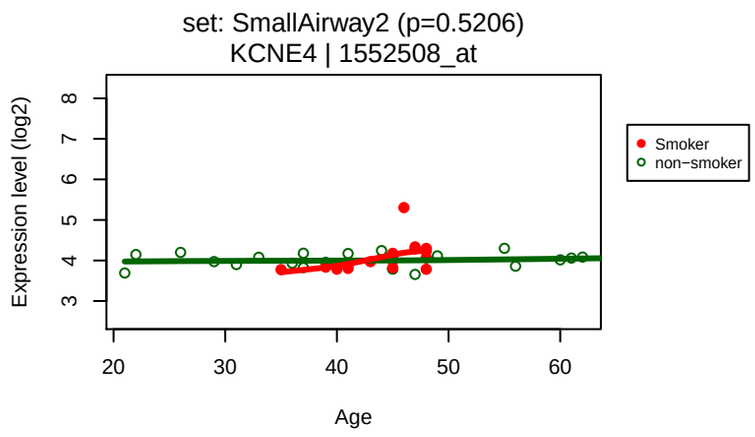
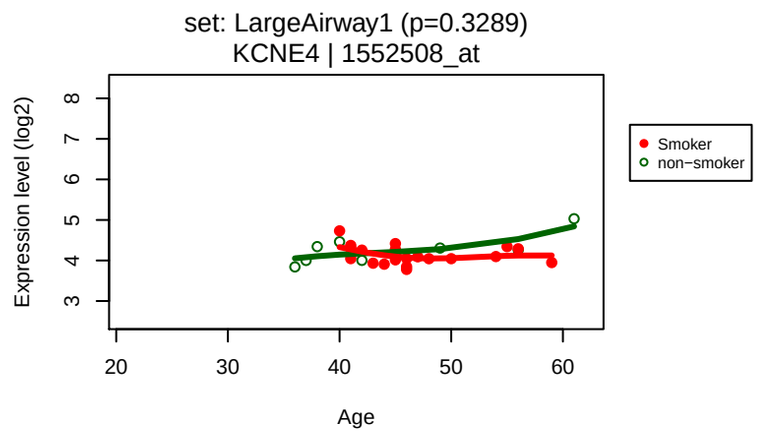

Gene expression of smokers and nonsmokers
gene: KCNE4 | 1552508_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1118
p=0.5206
p=0.3873
p=0.3289
p=0.5141
p=0.0107
p=0.86
p=0.1125
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

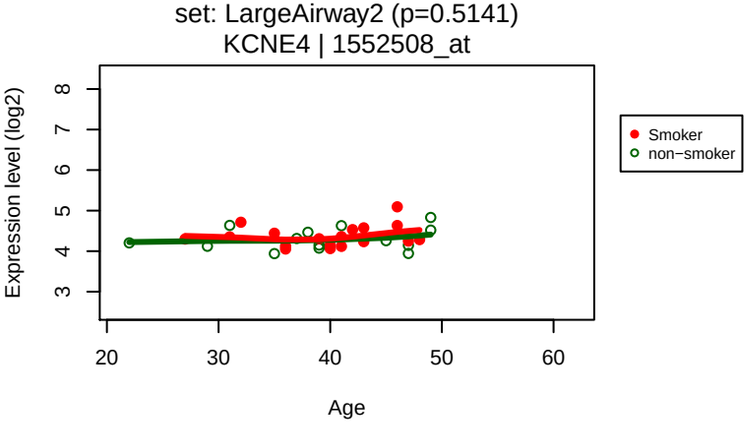

Gene expression ratio (smokers/nonsmokers)
gene: KCNE4 | 1552508_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1118
p=0.5206
p=0.3873
p=0.3289
p=0.5141
p=0.0107
p=0.86
p=0.1125
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

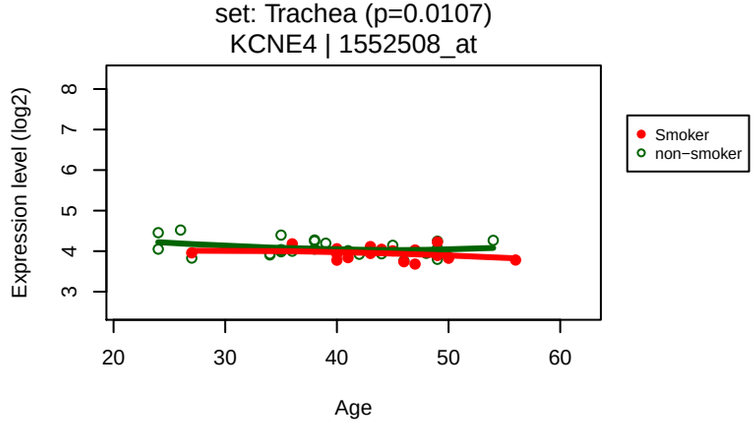

Gene expression ratio in Males
gene: KCNE4 | 1552508_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5735
p=0.584
p=0.0089
p=0.1503
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNE4 | 1552508_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.31
p=0.156
p=0.5456
p=0.5945
SmallAirway2
LargeAirway2
Trachea
Macrophage2

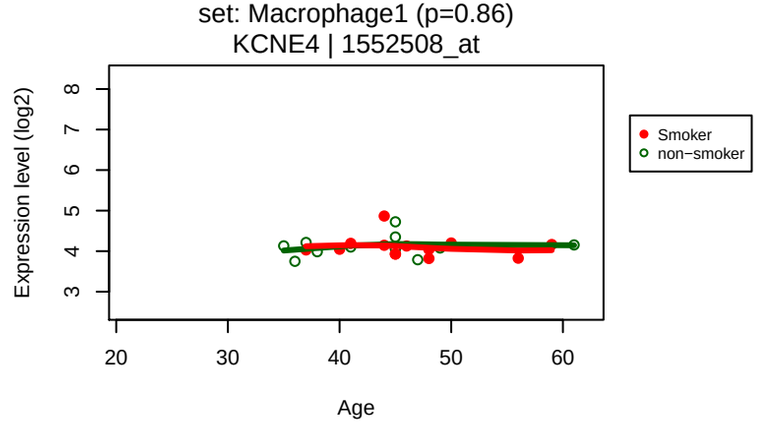

set: SmallAirway1 (p=0.1118)
KCNE4 | 1552508_at
Expression level (log2)
Age
Smoker
non−smoker

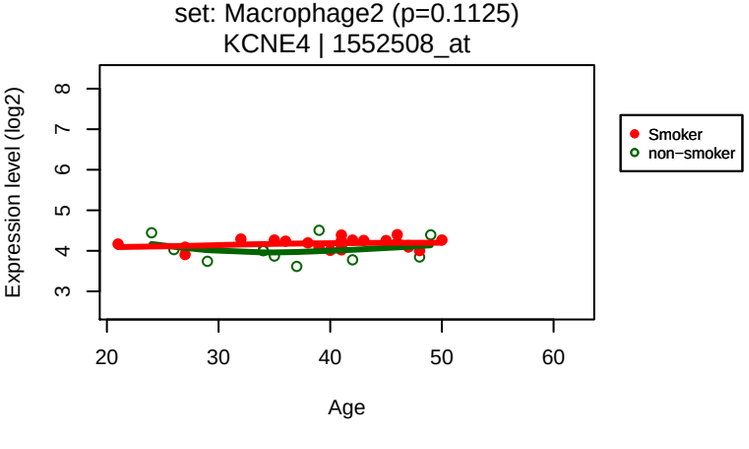

set: SmallAirway2 (p=0.5206)
KCNE4 | 1552508_at
Expression level (log2)
Age
Smoker
non−smoker

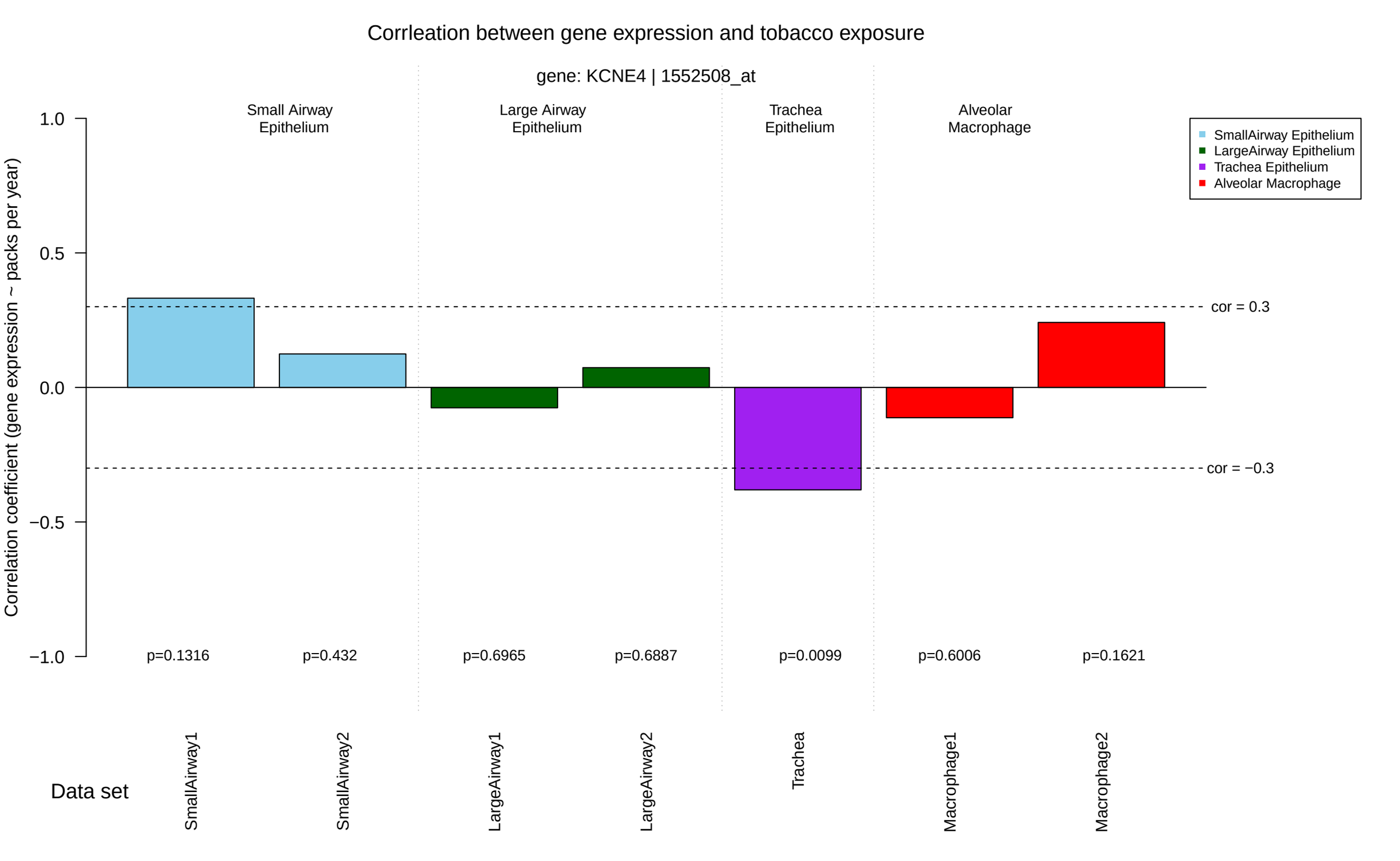

set: LargeAirway1 (p=0.3289)
KCNE4 | 1552508_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5141)
KCNE4 | 1552508_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0107)
KCNE4 | 1552508_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.86)
KCNE4 | 1552508_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1125)
KCNE4 | 1552508_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNE4 | 1552508_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1316
p=0.432
p=0.6965
p=0.6887
p=0.0099
p=0.6006
p=0.1621
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2